



Semester : Spring/Summer/Fall Year : 2026

Course Title : Computer Graphics
Course Code : CSE411
Assignment On : Bresenham's Line Drawing Algorithm & Mid-Point Circle Drawing Algorithm

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Date of Submission: 01-07-2026

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Assignment

Solution: Bresenham's Line Drawing Algorithm

Points: (6,6), (8,2), (12,6)

Line 1: (6,6) $\rightarrow$ (8,2)

$$m = \frac{\Delta y}{\Delta x} = \left| \frac{6-2}{8-6} \right| = \frac{4}{2} = 2 > 1 \quad \therefore y \text{ drives}$$

$$\therefore P_0 = 2\Delta x - \Delta y = 2 \times 2 - 4 = 0$$

$$x \text{ change} = +1, y \text{ change} = -1$$

Here,

if $P_k < 0$ then x same, $y + \text{change}$, $P_{k+1} = P_k + 2\Delta x$

if $P_k \geq 0$ then $x + \text{change}$, $y + \text{change}$, $P_{k+1} = P_k + 2\Delta x - 2\Delta y$

k	P_0	P_{k+1}	x_{k+1}	y_{k+1}
0			6	6
1	0	-4	7	5
2	-4	0	7	4
3	0	-4	8	3
4	-4		8	2

$\therefore$ Drawing Pixels are: (6,6), (7,5), (7,4), (8,3), (8,2)

Line 2: (8,2) $\rightarrow$ (12,6)

$$\Delta x = |12-8| = 4, \Delta y = |6-2| = 4 \quad \therefore m = \frac{\Delta y}{\Delta x} = \frac{4}{4} = 1 \quad \therefore x \text{ drives}$$

$$\therefore P_0 = 2\Delta y - \Delta x = 2 \times 4 - 4 = 4, x \text{ change} = +1, y \text{ change} = +1$$

Here,

if $P_k < 0 \rightarrow x + \text{change}, y \text{ same}, P_{k+1} = P_k + 2\Delta y$

if $P_k \geq 0 \rightarrow x + \text{change}, y + \text{change}, P_{k+1} = P_k + 2\Delta y - 2\Delta x$

k	P ₀	P _{k+1}	x _{k+1}	y _{k+1}
0			8	2
1	4	4	9	3
2	4	4	10	4
3	4	4	11	5
4	4	4	12	6

∴ Drawing Pixels are: (8,2), (9,3), (10,4), (11,5), (12,6)

Line 03: (6,6) → (12,6)

$$\Delta x = |6-12| = 6, \Delta y = |6-6| = 0, m = \frac{\Delta y}{\Delta x} = \frac{0}{6} = 0 \leq 1 \therefore x \text{ drives}$$

$$P_0 = 2\Delta y - \Delta x = 2 \times 0 - 6 = -6, x \text{ change} = +1, y \text{ change} = 0$$

Hence,

if $P_k < 0 \rightarrow x + \text{change}, y \text{ same}, P_{k+1} = P_k + 2\Delta y$

if $P_k \geq 0 \rightarrow x + \text{change}, y + \text{change}, P_{k+1} = P_k + 2\Delta y - 2\Delta x$

k	P ₀	P _{k+1}	x _{k+1}	y _{k+1}
			6	6
	-6	-6	7	6
	-6	-6	8	6
	-6	-6	9	6
	-6	-6	10	6
	-6	-6	11	6
	-6	-6	12	6

∴ Drawing Pixels are: (6,6), (7,6), (8,6), (9,6), (10,6), (11,6), (12,6)

Mid Point Circle Drawing Algorithm:

Assume that, center $(0,0)$, Given that: radius = 4 (for all circles)

$\therefore$ Initial values: $x=0, y=4 \therefore P_0 = 1 - r = 1 - 4 = -3$

Hence,

if $P_k < 0 \rightarrow x+1, y$ same, $P_{k+1} = P_k + 2x_{k+1} + 1$

if $P_k \geq 0 \rightarrow x+1, y-1, P_{k+1} = P_k + 2x_{k+1} - 2y_{k+1} + 1$

Table (1st Octant, Origin-based):

k	P_0	P_{k+1}	x_{k+1}	y_{k+1}
0			0	4
1	-3	0	1	4
2	0	-1	2	3
3	-1	6	3	3
4	6	stop		

$\therefore$ 1st Octant Points (Origin): $(0,4), (1,4), (2,3), (3,3)$

8-way Symmetry (Origin-based full circle):

from (x,y) to $(\pm x, \pm y), (\pm y, \pm x)$

Origin Point	All 8 Symmetric Points
$(0,4)$	$(0,4), (0,-4), (4,0), (-4,0)$
$(1,4)$	$(1,4), (-1,4), (1,-4), (-1,-4), (4,1), (-4,1), (4,-1), (-4,-1)$
$(2,3)$	$(2,3), (-2,3), (2,-3), (-2,-3), (3,2), (-3,2), (3,-2), (-3,-2)$
$(3,3)$	$(3,3), (-3,3), (3,-3), (-3,-3)$

P.T.O

Shift to each center

(i) Purple circle - center (7,8)

Adding (7,8) to all origin points:

Origin	Purple Point	Origin	Purple Point	Origin	Purple Point
(0,4)	(7,12)	(0,-4)	(7,4)	(4,0)	(11,8)
(-4,0)	(3,8)	(1,4)	(8,12)	(-4,-4)	(8,4)
(-1,4)	(6,12)	(-1,-4)	(6,4)	(4,1)	(11,9)
(4,-1)	(11,7)	(-4,1)	(3,9)	(-4,-1)	(3,7)
(2,3)	(9,11)	(2,-3)	(9,5)	(-2,3)	(5,11)
(-2,-3)	(5,5)	(3,2)	(10,10)	(3,-2)	(10,6)
(-3,2)	(4,10)	(-3,-2)	(4,6)	(3,3)	(10,11)
(3,-3)	(10,5)	(-3,3)	(4,11)	(-3,-3)	(4,5)

(ii) Red circle - center (8,8)

Adding (8,8) to all origin points:

Origin	Red Point	Origin	Red Point	Origin	Red Point
(0,4)	(8,12)	(0,-4)	(8,4)	(4,0)	(12,8)
(-4,0)	(4,8)	(1,4)	(9,12)	(1,-4)	(9,4)
(-1,4)	(7,12)	(-1,-4)	(7,4)	(4,1)	(12,9)
(4,-1)	(12,7)	(-4,1)	(4,9)	(-4,-1)	(4,7)
(2,3)	(10,11)	(2,-3)	(10,5)	(-2,3)	(6,11)
(-2,-3)	(6,5)	(3,2)	(11,10)	(3,-2)	(11,6)
(-3,2)	(5,10)	(-3,-2)	(5,6)	(3,3)	(11,11)
(3,-3)	(11,5)	(-3,3)	(5,11)	(-3,-3)	(5,5)

iii) Black Circle - center (11,8)

Adding (11,8) to all origin points

Origin	Black Point	Origin	Black Point	Origin	Black Point
(0,4)	(11,12)	(0,-4)	(11,4)	(4,0)	(15,8)
(-4,0)	(7,8)	(11,4)	(12,12)	(11,-4)	(12,4)
(-1,4)	(10,12)	(-1,-4)	(10,4)	(4,1)	(15,9)
(4,-1)	(15,7)	(4,1)	(7,9)	(-4,-1)	(7,7)
(2,3)	(13,11)	(2,-3)	(13,5)	(-2,3)	(9,11)
(-2,-3)	(9,5)	(3,2)	(14,10)	(3,-2)	(14,6)
(-3,2)	(8,10)	(3,-2)	(8,6)	(-3,3)	(14,11)
(3,-3)	(14,5)	(-3,3)	(8,11)	(-3,-3)	(8,5)

Trimming:

Purple & Red circle are overlapping. So trimming the inner overlapping portion. No need. No need to change red & black circle.

$$\therefore \text{Boundary} = (7+8)/2 = 7.5$$

So, Purple: $x \leq 7$ are allowed (Reject if $x \geq 8$)

Red: $x \geq 8$ are allowed (Reject if $x \leq 7$)

$\therefore$ Purple Final Points ($x \leq 7$ only):

(7,12), (7,4), (3,8), (6,12), (6,4), (3,9), (3,7), (5,11), (5,5), (4,10), (4,6), (4,11), (4,5)

$\therefore$ Red Final Points ($x \geq 8$ only):

(8,12), (8,4), (12,8), (9,12), (9,4), (12,9), (12,7), (10,11), (10,5), (11,10), (11,6), (11,11), (11,5)

$\therefore$ Black Final Points (No need to trim):

(11,12), (11,4), (15,8), (7,8), (12,12), (12,4), (10,12), (10,4), (15,9), (15,7), (7,9), (7,7), (13,11), (13,5), (9,11), (9,5), (14,10), (14,6), (8,10), (8,6), (14,11), (14,5), (8,11), (8,5)